Workshop Pole Barn: Materials and Cost Estimate

Bill of materials with Home Depot, Lowe's, and specialty-supplier ballpark prices (Mountain West, 2026) · 32 × 48 ft · planned for 2028

Source codes: **HD** Home Depot · **LO** Lowe's · **SP** specialty supplier · **Q** vendor quote required. Every quantity is computed from the building's dimensions in `docs/workshop_model.js`, and every total is the quantity times the unit price, summed. Prices are 2026 ballpark ranges, not quotes.

This is a planning estimate, not a fixed-price bid

Prices are 2026 ballpark ranges for the Treasure Valley and Boise area. Lumber moves with commodity prices; steel and trusses must be quoted by your supplier with current numbers. The **Midpoint** column is halfway between each line's low and high, so it is a planning number, not a prediction.

Items marked **Q** **must be quoted**: the pre-fabricated trusses, the steel package, the laminated columns, and the wind-rated overhead door together are roughly 40–50% of the materials cost, and prices vary a lot by supplier. The truss price in particular is a placeholder until quotes arrive.

Summary by Phase

#	Phase / category	Low	Midpoint	High
1	Site Work & Slab Sub-Base	$3,672	**$4,399**	$5,125
2	Concrete (Piers, Slab, Apron, Deck Footings)	$10,868	**$12,363**	$13,857
3	Foundation Hardware	$2,648	**$3,564**	$4,480
4	Laminated Posts (3-ply 2×8) **Q**	$4,352	**$5,668**	$6,984
5	Pre-Engineered Gambrel Attic Trusses **Q**	$11,824	**$15,339**	$18,853
6	Wall Framing (Girts & Blocking)	$1,862	**$2,200**	$2,537
7	Roof Framing (Purlins & Roof Blocking)	$1,128	**$1,310**	$1,491
8	Sheathing	$3,474	**$4,182**	$4,890
9	Weather Barrier & Underlayments	$2,493	**$2,815**	$3,137
10	Rainscreen (Furring & Flashings)	$1,389	**$1,679**	$1,968
11	Steel Package (Roof + Walls + Trim) **Q**	$10,084	**$12,760**	$15,436
12	Splash Board	$405	**$463**	$521
13	Doors & Windows **Q**	$7,700	**$10,935**	$14,170
14	Loft Floor Decking	$1,681	**$2,098**	$2,515
15	Interior Loft Stairs (L-shape)	$825	**$1,051**	$1,277
16	Landing Deck (14′ × 10′, free-standing)	$1,306	**$1,561**	$1,815
17	Hoist Beam, Braces & Trolley	$885	**$1,269**	$1,653
18	Electrical	$4,235	**$5,802**	$7,369
19	HVAC (Mini-Split + Wood Stove)	$4,220	**$6,185**	$8,150
20	Insulation (Phase 2 — when conditioning)	$3,569	**$4,492**	$5,415
21	Fasteners & Miscellaneous Hardware	$724	**$945**	$1,165
MATERIALS TOTAL (excludes labor, permits, and utility hookups)		**$79,344**	$101,076	**$122,808**

Not in the materials total

- **Labor.** Owner-build assumes you provide it; a hired contractor adds 30–50%.

- **Permits, plan review, and engineering**, **pier holes and dirt work**, **the power feeder trench and connection** (the specs call for no water or septic), **equipment rental**, and **sales tax**: these are in the budget table at the end of this page.
- **Solar, generator, and batteries** are an owner-future item and are not in either table.

1 · Site Work & Slab Sub-Base

Item	Qty	Unit	Unit low	Unit high	Total low	Total high	Src	Notes
¾″ minus crushed gravel base, compacted 4″	19	cu yd	$28	$45	$532	$855	SP	Local gravel pit — much cheaper than HD bagged. It goes on the compacted pad fill.
10 mil black poly vapor barrier 20×100	1	roll	$130	$180	$130	$180	HD	Husky brand, sealed seams with poly tape.
4″ XPS rigid foam 4×8 (R-20 under slab)	48	sheet	$60	$80	$2,880	$3,840	HD	1,536 sq ft slab area. Owens Corning Foamular 250 or equiv.
Erosion control / silt fence	1	lot	$80	$150	$80	$150	HD	Per permit.
Layout supplies (stakes, mason line, marking paint)	1	lot	$50	$100	$50	$100	HD	—
PHASE SUBTOTAL					**$3,672**	**$5,125**		

Cost-cut options here: drop under-slab foam to 2″ (R-10) saves ~$1,400; skip foam entirely if you will never heat above 50 °F (saves ~$2,900 but you lose comfort and the slab will sweat in summer).

2 · Concrete (Piers, Slab, Apron, Deck Footings)

Item	Qty	Unit	Unit low	Unit high	Total low	Total high	Src	Notes
Ready-mix 4,000 psi w/ fiber mesh	40	cu yd	$170	$210	$6,800	$8,400	SP	Includes ~26 yd slab + 9 yd piers + 3 yd apron + 2 yd deck footings.
Short-load surcharge (2 trucks)	2	trip	$75	$150	$150	$300	SP	Likely needed for pier-pour + slab-pour separately.
Sonotube 18″Ø × 12 ft (building piers)	19	ea	$80	$100	$1,520	$1,900	HD	Cut to length on site. One 12 ft tube cuts into two piers, so ask whether half as many will do.
Sonotube 18″Ø × 12 ft (deck piers)	4	ea	$80	$100	$320	$400	HD	Landing deck piers; the engineer sizes them.
#4 rebar 20 ft (slab + piers + apron)	155	stick	$10	$13	$1,550	$2,015	HD	Slab grid 16″ OC EW = ~120 sticks. Piers ~18, apron ~17. Buy ahead, cut on site.
Rebar chairs / plastic bolsters (bag of 50)	9	bag	$12	$18	$108	$162	HD	To hold slab rebar at mid-depth.
Rebar tie wire + tie tool / pliers	1	lot	$40	$80	$40	$80	HD	—
Concrete sealer (Foundation Armor or equiv)	2	5-gal	$130	$200	$260	$400	HD	Penetrating sealer for slab. Apply after 28-day cure.
Saw-cut control joints (rental + blades)	1	day	$120	$200	$120	$200	HD	Cut joints in a pattern of about 11 ft panels within 12 hours of pour.
PHASE SUBTOTAL					**$10,868**	**$13,857**		

3 · Foundation Hardware

Item	Qty	Unit	Unit low	Unit high	Total low	Total high	Src	Notes
Sturdi-Wall Plus / Perma-Column wet-set bracket (sized for 3-ply 2×8)	19	ea	$130	$220	$2,470	$4,180	SP	Order from Perma-Column or Mid-States. Set in wet concrete during pier pour.

Item	Qty	Unit	Unit low	Unit high	Total low	Total high	Src	Notes
Deck post bases (Simpson ABU, hot-dipped, 6×6 size)	4	ea	$32	$45	$128	$180	HD	Wet-set in the deck pier pour; sized for the 6×6 deck legs.
Galvanized anchor bolts / threaded rod stock	1	lot	$50	$120	$50	$120	HD	Bracket hardware usually included; this is overage.
PHASE SUBTOTAL					**$2,648**	**$4,480**		

4 · Laminated Posts (3-ply 2×8)

Item	Qty	Unit	Unit low	Unit high	Total low	Total high	Src	Notes
3-ply 2×8 SPF laminated PT column, 16 ft	19	ea	$224	$360	$4,256	$6,840	Q	Pre-fab laminated from Perma-Column, Mid-States, or local truss supplier. Each post runs from its bracket, just above the pad surface, to the truss carrier at the 14 ft eave. Alternative: site-build with 3-ply 2×8 #2 SPF (bottom 4 ft PT for splash), full glue + nail per ICC ESR. Cost-cut: site-build saves ~$1,200 but takes 2 days labor.
Construction adhesive (PL Premium, if site-laminating)	12	tube	$8	$12	$96	$144	HD	Only if site-fabricating posts.
PHASE SUBTOTAL					**$4,352**	**$6,984**		

5 · Pre-Engineered Gambrel Attic Trusses

Item	Qty	Unit	Unit low	Unit high	Total low	Total high	Src	Notes
Common gambrel attic truss, 32′ span, 2′ OC, PE stamped (120 psf ground snow, 40 psf loft floor)	23	ea	$450	$700	$10,350	$16,100	Q	Quote 3 suppliers: Western Truss (Caldwell), BMC Truss (Boise), Builders FirstSource. State explicitly: 120 psf ground snow (the county's design load here), 115 mph wind Exp C, 40 psf loft LL, gambrel attic with finished room space. The unit price is a placeholder: a 32 ft attic truss for a loft floor and this snow load will likely price above an ordinary roof truss.
Gable-end truss (drop-in)	2	ea	$300	$500	$600	$1,000	Q	Solid-faced for siding nailers; outlooks for gable overhang.
Truss delivery + crane / boom truck	1	trip	$400	$1,000	$400	$1,000	Q	Long-bed delivery + boom set is faster & safer than rolling in by hand.
Truss-clip hurricane ties (Simpson H2.5A, box of 100)	1	box	$85	$110	$85	$110	HD	2 per truss bearing × 25 trusses × 2 sides = 100 ties needed.
Truss-set blocking (2×6 SPF, 8 ft)	27	stick	$7	$9	$189	$243	HD	Between trusses at heel + ridge for stability.
Permanent lateral bracing per truss design	1	lot	$200	$400	$200	$400	HD	2×4 SPF + framing nails per truss-design BCSI sheets.
PHASE SUBTOTAL					**$11,824**	**$18,853**		

> **Critical:** truss order needs **4–8 week lead time**. Get the quote & place the deposit BEFORE you start the foundation so trusses arrive ready when walls are up.

6 · Wall Framing (Girts & Blocking)

Item	Qty	Unit	Unit low	Unit high	Total low	Total high	Src	Notes
2×6×16 SPF #2 (girts: north, south, and west walls)	48	stick	$13	$16	$624	$768	HD	Laid flat on the outside face of the posts and screwed to each one: 6 rows above the base girt at 24″ OC. Every joint lands on a post, and neighboring rows start with boards of different lengths so their joints fall on different posts. The pieces cut out at the windows and the walk door make the sills and jack studs.
2×6×18 SPF #2 (east wall girts, each cut in half)	6	stick	$17	$22	$102	$132	HD	The door jambs leave a 9 ft run on each side of the door in every row, so one 18 ft board makes both pieces with no waste. Not every yard stocks 18 footers: ask.
2×6×16 PT (base girts, ground contact)	10	stick	$20	$26	$200	$260	HD	The lowest row, its bottom edge at the floor line (13″ above the pad). It is pressure-treated because it is the row closest to the ground and the slab, and it backs the lower edge of the sheathing. It stops at the doors, so each of the east wall's two 9 ft pieces takes a board of its own. The splash board below it (phase 12) is screwed to the posts.
3½″ structural screws, girts to posts (100 ct)	5	box	$55	$80	$275	$400	HD	457 screws: 3 at each post a board crosses and 2 at each board end. The county's minimum is four 20d nails per girt at each post, so the engineer confirms the screw schedule.
2×6×8 SPF #2 (blocking, headers)	32	stick	$7	$9	$224	$288	HD	For door jacks, headers, blocking, gable infill.
2×6×8 SPF #2 (window sills and jack studs)	11	stick	$7	$9	$77	$99	HD	7 windows, about 81 lin ft. The girt above a sidewall window is its header, so each opening needs only a sill and two jack studs once the girt row across it is cut. Windows sit between posts, so the posts carry the wall.
(2) 2×10 LVL header, 18 ft (over the 14×12 OH door)	2	ea	$120	$200	$240	$400	HD	LP / GP LVL 1¾×9¼. Placeholder: the specs call for (3) 2×12 LVL or a steel beam sized by the engineer, no deeper than 12 in, because it must fit in the 14 in between the door head and the underside of the loft floor. Truss bearing above goes onto the LVL via post-jacks.
(2) 2×10 LVL header, 8 ft (loft doors + walk door)	2	ea	$60	$95	$120	$190	HD	Loft door header + walk door header.
PHASE SUBTOTAL					**$1,862**	**$2,537**		

7 · Roof Framing (Purlins & Roof Blocking)

Item	Qty	Unit	Unit low	Unit high	Total low	Total high	Src	Notes
2×6×16 SPF #2 (purlins on top of truss chords)	66	stick	$13	$16	$858	$1,056	HD	~1,040 lin ft purlins on 4 slopes at 24″ OC.
Simpson H1 / TC clips (purlin to truss)	2	box of 100	$70	$95	$140	$190	HD	1 clip per purlin-truss intersection ~200 clips.
OSB H-clips for roof deck (bag of 250)	1	bag	$30	$45	$30	$45	HD	1 per panel edge between purlins.
Ridge / gambrel-break blocking (2×6 cut to suit)	1	lot	$100	$200	$100	$200	HD	For ridge cap nailing + steel termination at pitch break.
PHASE SUBTOTAL					**$1,128**	**$1,491**		

8 · Sheathing

Item	Qty	Unit	Unit low	Unit high	Total low	Total high	Src	Notes
½″ OSB 4×8 sheathing (walls + gable infill)	82	sheet	$18	$26	$1,476	$2,132	HD	~2,380 sq ft net wall + gable area, +10% waste. APA Exposure 1.
⅝″ OSB 4×8 roof deck	73	sheet	$26	$36	$1,898	$2,628	HD	~2,100 sq ft roof (with a 1 ft eave and gable overhang) + 10% waste. APA Exposure 1.
8d ring-shank framing nails (30 lb box)	2	box	$50	$65	$100	$130	HD	For OSB to framing.
PHASE SUBTOTAL					**$3,474**	**$4,890**		

9 · Weather Barrier & Underlayments

Item	Qty	Unit	Unit low	Unit high	Total low	Total high	Src	Notes
Tyvek CommercialWrap 9′ × 100′	4	roll	$230	$280	$920	$1,120	HD	2,380 wall sq ft, +20% lap = 2,900 sq ft, 900/roll. CommercialWrap is the more durable version & appropriate for an unconditioned shop.
Tyvek tape (1.88″ × 165′)	4	roll	$22	$28	$88	$112	HD	Seam & flashing tape — buy Tyvek brand, not generic.
Grace Ice & Water Shield 36″ × 66.7′ (200 sq ft)	9	roll	$115	$145	$1,035	$1,305	HD	Full lower slope (~1,160 sq ft) + pitch break + penetrations. Most critical detail for a snow climate.
Synthetic underlayment 48″ × 250′ (1,000 sq ft)	3	roll	$130	$170	$390	$510	HD	Titanium UDL30, Grip-Rite, etc. Full roof, with laps.
Plastic cap nails for underlayment (5 lb box)	2	box	$30	$45	$60	$90	HD	—
PHASE SUBTOTAL					**$2,493**	**$3,137**		

10 · Rainscreen (Furring & Flashings)

Item	Qty	Unit	Unit low	Unit high	Total low	Total high	Src	Notes
1×4×8 PT furring strips	193	stick	$5	$7	$965	$1,351	HD	80 furring runs at 16 ft each = 160 sticks for walls (use 2 per 16 ft run). Plus gables ~33 sticks.
Insect screen / drip-edge strip (Cor-A-Vent SV-3 or equiv)	8	10 ft pc	$18	$26	$144	$208	SP	At base of walls + eave vent. Buy from siding distributor.
Aluminum drip-edge flashing (10 ft)	16	stick	$10	$14	$160	$224	HD	~160 lin ft for splash board top, +bug screen.
Coil stock for window/door flashings (8″ × 50′)	1	roll	$60	$95	$60	$95	HD	Color-matched coil for custom bends.
2½″ stainless ring-shank nails for furring (5 lb box)	2	box	$30	$45	$60	$90	HD	SS prevents tannin streaks through PT.
PHASE SUBTOTAL					**$1,389**	**$1,968**		

11 · Steel Package (Roof + Walls + Trim)

Item	Qty	Unit	Unit low	Unit high	Total low	Total high	Src	Notes
26 ga steel roof panel (beige w/ Dripstop)	2,100	sq ft	$1.90	$2.90	$3,990	$6,090	Q	Bare-roll prices; cut to exact slope length.
26 ga steel wall panel (barn red)	2,380	sq ft	$1.80	$2.70	$4,284	$6,426	Q	Bare-roll prices; vertical orientation, cut to full wall height.

Item	Qty	Unit	Unit low	Unit high	Total low	Total high	Src	Notes
Trim package — corner, base, eave, gambrel-break, ridge cap, J-channel	1	pkg	$900	$1,500	$900	$1,500	Q	Have supplier itemize; ridge cap MUST be vented for moisture relief.
Snow guards (twin-rail or pad-style)	20	ea	$10	$20	$200	$400	SP	On the lower slope over the walk door and any other doorway under a roof edge. The east gable has no roof slope above it. SnowTrapper, S-5! brand.
#10 × 1½" hex roofing screws w/ EPDM (250 ct, color-matched)	14	box	$40	$55	$560	$770	HD	Two colors needed (beige + red). Calculated at about ¾ screw per sq ft of panel.
Closure strips (foam, inside & outside profile)	1	lot	$150	$250	$150	$250	Q	At ridge cap and eaves — keeps wind-driven snow out.
PHASE SUBTOTAL					**$10,084**	**$15,436**		

Get an itemized package quote from a regional steel supplier (e.g. *Idaho Steel Products – Caldwell, Boise Steel Roofing, R-Panel/Metal Sales – Twin Falls*). Buying loose at HD/Lowe's costs ~50–80% more. Specify: 26 ga / Galvalume painted / 2-color (beige roof + barn red walls) / factory Dripstop on roof / 50-year paint warranty / vented ridge cap.

12 · Splash Board

Item	Qty	Unit	Unit low	Unit high	Total low	Total high	Src	Notes
2×10×16 PT (.40 or .60 ground contact)	12	stick	$30	$38	$360	$456	HD	160 lin ft perimeter at base of wall (less the door openings), lapped over piers.
HD hot-dipped galvanized 3" deck screws (5 lb box)	1	box	$45	$65	$45	$65	HD	Fasten splash board to posts/blocking. NEVER use bright/EG with PT.
PHASE SUBTOTAL					**$405**	**$521**		

13 · Doors & Windows

Item	Qty	Unit	Unit low	Unit high	Total low	Total high	Src	Notes
14′ × 12′ insulated sectional OH door, wind-rated, low-headroom track (Wayne Dalton 9100 / Clopay 4053)	1	ea	$3,500	$6,500	$3,500	$6,500	Q	Wind rated to ≥115 mph + insulated. Get an installed quote. The camper is 10′-11" tall, so a 12′ door clears it by 13 in. The loft floor leaves 14 in above the door, so ask for a low-headroom track kit that needs 12 in or less.
LiftMaster 8500W jackshaft opener (wall-mount, side-mount)	1	ea	$450	$650	$450	$650	SP	A wall-mount opener keeps the space under the loft floor free for the low-headroom track.
3070 insulated steel walk door, pre-hung w/ frame	1	ea	$400	$800	$400	$800	HD	9-lite or solid. Heavy-duty lockset extra.
5′ × 6′ double exterior insulated door (loft, hinged out OR sliding barn-door)	1	set	$1,200	$2,200	$1,200	$2,200	Q	Spec: weather-sealed, locking from inside, opens fully clear for hoist. Custom shop door OR pair of 30" exterior pre-hung.
Insulated vinyl window, 4′ × 3′ slider, low-E, DP-30 or better, with jamb extensions	6	ea	$260	$480	$1,560	$2,880	HD	Nail-fin windows in the north and south walls, sill 5′ above the floor. Ask for the design-pressure (DP) rating the engineer gives for 115 mph, and order jamb extensions deep enough to reach the inside face of the posts, 9¼" behind the face of the sheathing.
Insulated vinyl window, 3′ × 4′ single-hung, low-E, DP-30 or better (west loft wall)	1	ea	$240	$440	$240	$440	HD	Centered in the west gable, sill 3′ above the loft floor, so it lights the loft and the stair arrival.

Item	Qty	Unit	Unit low	Unit high	Total low	Total high	Src	Notes
Window flashing: self-adhered sill pans and flashing tape	1	lot	$60	$120	$60	$120	HD	Sill pan, then jamb tape, then head tape, lapped under the wrap above and over the wrap below.
Steel J-trim and head flashing for the windows	7	ea	$20	$40	$140	$280	SP	Color-matched; ask the steel supplier to cut it with the trim package.
Door hardware — locks, dead-bolts, weather strip, exterior pulls	1	lot	$150	$300	$150	$300	HD	—
PHASE SUBTOTAL					**$7,700**	**$14,170**		

14 · Loft Floor Decking

Item	Qty	Unit	Unit low	Unit high	Total low	Total high	Src	Notes
¾″ T&G Sturd-I-Floor OSB 4×8 (Advantech preferred)	51	sheet	$30	$45	$1,530	$2,295	HD	1,536 sq ft floor + 5% waste = 1,610 sq ft. Advantech ($45) much better moisture resistance during build.
Subfloor adhesive (PL400, 28 oz)	13	tube	$7	$10	$91	$130	HD	Glue + screw — silences floor squeaks.
2½″ deck screws for subfloor (5 lb box)	2	box	$30	$45	$60	$90	HD	—
PHASE SUBTOTAL					**$1,681**	**$2,515**		

15 · Interior Loft Stairs (L-shape)

Item	Qty	Unit	Unit low	Unit high	Total low	Total high	Src	Notes
2×12×12 SPF #1 (stringers, first flight)	3	stick	$40	$55	$120	$165	HD	11 risers, 6′-5″ up over 9′-2″ of run, to the landing.
2×12×14 SPF #1 (stringers, second flight)	3	stick	$47	$64	$141	$192	HD	13 risers, 7′-7″ up over 11′-0″ of run, to the loft floor.
2×10×12 SPF (treads or sub-treads)	12	stick	$22	$30	$264	$360	HD	2 treads per stick. ~24 treads total.
2×4×8 SPF (handrail + landing frame)	10	stick	$4	$6	$40	$60	HD	—
Pre-fab oak/poplar handrail + balusters (kit)	1	kit	$200	$400	$200	$400	HD	Optional — code requires graspable handrail 34–38″ above nosing.
LSC stringer connectors + joist hangers	1	lot	$60	$100	$60	$100	HD	Simpson LSC adjustable stringer connector.
PHASE SUBTOTAL					**$825**	**$1,277**		

16 · Landing Deck (14′ × 10′, free-standing)

Item	Qty	Unit	Unit low	Unit high	Total low	Total high	Src	Notes
6×6×16 PT posts (deck legs, full height to ground)	4	stick	$55	$75	$220	$300	HD	6×6 for lateral stability: the deck is 14 ft tall, so it needs the bigger leg. 16 ft is the stock length; each leg is cut to about 14 ft.
2×10×16 PT (beams & outer joists)	4	stick	$30	$38	$120	$152	HD	Two doubled beams east-west on the post lines.
2×10×14 PT (joists @ 12″ OC)	10	stick	$25	$32	$250	$320	HD	Run north-south between the beams. Tighter spacing for snow load.
5/4×6×12 PT decking	26	stick	$10	$14	$260	$364	HD	140 sq ft + waste. Alt: composite for ~3× cost.
2×6×8 PT (diagonal knee-braces, 4 needed)	4	stick	$8	$11	$32	$44	HD	Critical for lateral stability — a 14 ft tall deck WILL sway without these.

Item	Qty	Unit	Unit low	Unit high	Total low	Total high	Src	Notes
2×4×8 PT (railing top + bottom + balusters)	30	stick	$4	$6	$120	$180	HD	42″ railing required for fall protection > 30″.
Simpson joist hangers (LUS210, box of 25)	1	box	$60	$85	$60	$85	HD	HDG for PT contact.
Simpson hurricane ties for beam-to-post (H1)	8	ea	$3	$5	$24	$40	HD	—
HDG structural screws (Headlok / GRK RSS, 4½″, 100 ct)	2	box	$45	$65	$90	$130	HD	For beam to post connections.
HDG #9 × 3″ deck screws (5 lb box)	2	box	$35	$50	$70	$100	HD	Camo or face screws.
Flashing and sealer at the wall	1	lot	$60	$100	$60	$100	HD	The deck stands free of the wall (no ledger), so this is only for a brace tie-in if the engineer wants one.
PHASE SUBTOTAL					**$1,306**	**$1,815**		

17 · Hoist Beam, Braces & Trolley

Item	Qty	Unit	Unit low	Unit high	Total low	Total high	Src	Notes
W6×12 steel I-beam, 21′-6″ (braced at the tip by two struts, rated for 1,000 lb at the hook)	1	ea	$323	$501	$323	$501	SP	Local steel yard (Idaho Steel Boise / Triumph Steel), cut to length. 9 ft inside + 12′-6″ out past the wall, which is 2 ft beyond the deck's outer edge. The engineer picks the size; a W6×15 is the likely next step up.
6×6×12 PT (hoist braces: two raking struts)	2	ea	$41	$56	$82	$112	HD	From a steel shoe on the outer end of each outer deck beam to the tip of the hoist beam, each about 11′-9″ long. Steel tube is the alternative if the engineer prefers it.
Beam-tip end plate and brace shoes (fabricated, galvanized)	1	lot	$120	$260	$120	$260	SP	A plate on the end of the beam with ears for the two struts, which is also the trolley's outer stop, and a steel shoe on the end of each outer deck beam to seat a strut. A welder or the steel yard makes them.
Beam trolley (1-ton capacity, fits W6 flange)	1	ea	$60	$120	$60	$120	SP	Harbor Freight, Northern Tool, Grainger.
Chain fall hoist (1-ton, 25 ft lift)	1	ea	$140	$320	$140	$320	SP	The hook runs from the ground up to the beam, about 25 ft. A hand chain fall that long is a special order; an electric hoist is the alternative.
Beam inner end-stop hardware (welded angles or bolt-on)	1	lot	$30	$60	$30	$60	SP	Keeps the trolley from running off the inside end; the tip plate stops it at the outer end.
EPDM weather boot / flashing for beam penetration	1	ea	$40	$80	$40	$80	HD	Custom-made from flat EPDM + butyl tape. CRITICAL detail.
Anchor plate / bolt-through hardware (beam to truss)	1	lot	$50	$120	$50	$120	HD	Coordinate with truss engineer for attachment.
Blocking (2×6 + 2×8 between trusses to spread beam load)	1	lot	$40	$80	$40	$80	HD	—
PHASE SUBTOTAL					**$885**	**$1,653**		

Engineering check: have the truss manufacturer review the hoist-beam attachment, and the engineer size the beam, the two braces, and the deck posts and piers that carry them. A beam that reaches 12′-6″ past the wall with 1,000 lb at its tip makes a serious uplift load on the truss top chord inside; the truss may need an extra ply at that location or a reinforced web. The braces take much of the tip load down into the deck, so the deck is part of the hoist.

18 · Electrical

Item	Qty	Unit	Unit low	Unit high	Total low	Total high	Src	Notes
200 A main-lug sub-panel (Square D HOM, 42 space)	1	ea	$220	$380	$220	$380	HD	Square D Homeline or Eaton BR. Main-lug, fed from the pump house or the house (feed source to settle).
Breakers — 15/20 A (10 ea), 30 A (3: RV, compressor, mini-split), 50 A (1: welder)	14	ea	mixed	mixed	$200	$400	HD	15/20 A standard, 30 A for the RV outlet, the compressor, and the mini-split (check its nameplate), 50 A for the welder.
4/0 AL feeder cable, wet-rated, in conduit (per ft)	200	ft	$8	$13	$1,600	$2,600	HD	Pulled through the 2″ PVC conduit from the pump house (about 29 ft south) or the house. 200 ft is a generous allowance: check distance + voltage drop. Alt: 2/0 CU for shorter runs.
2″ Schedule 40 PVC conduit 10 ft (sleeve in/out of building)	6	stick	$14	$18	$84	$108	HD	Sweep + sleeve at panel entry.
12 AWG NM-B Romex (250 ft roll, yellow)	3	roll	$95	$140	$285	$420	HD	20 A general circuits.
14 AWG NM-B Romex (250 ft, white) — lighting	2	roll	$70	$110	$140	$220	HD	15 A lighting circuits only.
10 AWG NM-B 250 ft (30 A circuits)	1	roll	$160	$220	$160	$220	HD	For RV TT-30 outlet + mini-split.
6/3 AWG (welder 50 A 6-50R run)	50	ft	$5	$7	$250	$350	HD	—
20 A duplex outlets (15 ea) + 20 A GFCI (6 ea)	21	ea	mixed	mixed	$150	$280	HD	Mix standard + GFCI (where required).
NEMA 6-50R outlet (welder)	1	ea	$25	$40	$25	$40	HD	—
NEMA TT-30R outlet (RV)	1	ea	$25	$40	$25	$40	HD	Weather-resistant in WP cover.
Wall switches (single + 3-way)	12	ea	$3	$8	$36	$96	HD	—
Steel handy boxes / 4″ square boxes	40	ea	$2	$4	$80	$160	HD	Exposed conduit-mount.
EMT ½″ conduit + couplings + clamps (exposed runs)	1	lot	$300	$500	$300	$500	HD	Exposed conduit looks cleanest in shop.
LED high-bay light 150 W (5000 K, ~22,000 lm)	6	ea	$60	$150	$360	$900	HD	Lithonia / Hyperikon. Mount at loft floor level.
Exterior LED wall pack w/ photocell	3	ea	$40	$85	$120	$255	HD	Above OH door, walk door, deck.
Misc — wire nuts, lugs, ground rod & clamp, in-use covers	1	lot	$200	$400	$200	$400	HD	—
PHASE SUBTOTAL					**$4,235**	**$7,369**		

19 · HVAC (Mini-Split + Wood Stove)

Item	Qty	Unit	Unit low	Unit high	Total low	Total high	Src	Notes
Mini-split heat pump 2-zone (1× 12k + 1× 9k indoor, Mr Cool or Senville)	1	system	$2,000	$3,500	$2,000	$3,500	SP	DIY pre-charged lines = no HVAC trade required. Mr Cool DIY series. Shoulder-season heat + summer cool.
Wood stove (Drolet HT3000 / Englander 30-NC, EPA cert)	1	ea	$1,200	$2,800	$1,200	$2,800	HD	Hearth/clearance per UL listing — won't go on slab without insulated pad.
Class A double-wall insulated chimney 6″ (DuraVent DVL / Simpson Dura-Plus, kit)	1	kit	$700	$1,200	$700	$1,200	SP	Through-the-wall or through-the-roof — through-wall typically lower cost & safer for a tall building.
Stove hearth pad (UL-listed, R-1.6 min)	1	ea	$120	$250	$120	$250	HD	—
Stovepipe + storm collar + rain cap + thimble	1	lot	$200	$400	$200	$400	HD	—
PHASE SUBTOTAL					**$4,220**	**$8,150**		

20 · Insulation (Phase 2 — when conditioning)

Item	Qty	Unit	Unit low	Unit high	Total low	Total high	Src	Notes
R-23 fiberglass batts 5½" × 16 OC (kraft-faced)	26	bag	$45	$60	$1,170	$1,560	HD	Walls — install behind drywall/OSB interior finish. ~2,110 sq ft of wall area to insulate.
R-30 batts (loft floor between joists)	11	bag	$55	$75	$605	$825	HD	If loft floor is over unconditioned space — but ours is conditioned both sides, so OPTIONAL.
R-49 blown-in cellulose for upper loft ceiling (above flat upper-truss tie)	1	lot	$500	$900	$500	$900	SP	Trusses commonly have a flat tie above loft creating an attic; blow cellulose on top.
6 mil poly vapor barrier (interior side of walls)	2	roll	$80	$120	$160	$240	HD	Cold-climate zone 6B. Verify with builder — kraft-face on batts may suffice.
Interior finish — ½" OSB or T1-11 (full shop interior)	63	sheet	$18	$30	$1,134	$1,890	HD	Far more abuse-resistant than drywall in a shop. Alt: skip for now, install later. It is screwed to horizontal strapping across the posts, which this list does not price.
PHASE SUBTOTAL					**$3,569**	**$5,415**		

Phasing tip: build the shell first, conditioned later. The rainscreen + ½" OSB sheathing makes the shell fully weather-tight even with no insulation or interior finish. Move in equipment, work in it for a winter, decide which areas need heat — then phase in mini-split + insulation only where you actually use it.

21 · Fasteners & Miscellaneous Hardware

Item	Qty	Unit	Unit low	Unit high	Total low	Total high	Src	Notes
16d common nails (50 lb box)	1	box	$65	$90	$65	$90	HD	For framing.
Bag of 1¾" × 0.131 framing-nailer ring-shank coils	2	box	$80	$120	$160	$240	HD	If using gun.
3" structural screws (Headlok, 100 ct)	3	box	$50	$75	$150	$225	HD	Post-to-post and miscellaneous framing (the girt screws are in phase 6).
Construction adhesive (PL Premium, 28 oz)	12	tube	$7	$10	$84	$120	HD	Multi-use.
Caulks & sealants (DAP DynaFlex 230, butyl, NP1)	1	lot	$80	$150	$80	$150	HD	—
Hurricane / strap-tie variety pack	1	lot	$80	$150	$80	$150	HD	Simpson assortment.
Spray-on house wrap sealant (Stretch & Seal)	1	can	$25	$40	$25	$40	HD	For seams/corners.
Misc safety (gloves, glasses, ear plugs, N95)	1	lot	$80	$150	$80	$150	HD	—
PHASE SUBTOTAL					**$724**	**$1,165**		

Shopping Strategy and Sourcing

From Home Depot or Lowe's

- Dimensional lumber (2×4, 2×6, 2×8, 2×10): open accounts at both, and check prices weekly
- OSB sheathing (½″ and ⅝″): get the Pro-desk price and ask about bunk pricing
- PT lumber (1×4 furring, 2×10 splash board, 6×6 deck posts)
- Tyvek and Tyvek tape (Tyvek brand, not generic)
- Ice and water shield and synthetic underlayment
- Simpson connectors (hangers, hurricane ties, structural screws)
- Rigid foam (XPS) for under the slab
- Electrical: panel, breakers, wire, conduit, devices, LED fixtures
- The 3070 walk door (in stock or a one-week order)
- Concrete extras (rebar, Sonotube, sealer), caulk, adhesive, fasteners, and hardware

From a specialty or regional supplier

- **Trusses:** Western Truss (Caldwell), BMC, Builders FirstSource. 4–8 week lead time. PE stamp.
- **Laminated columns:** Perma-Column, Sturdi-Wall (Mid-States Building), Hansen Buildings, or site-build.
- **Steel package:** Idaho Steel Products (Caldwell), Boise Steel Roofing, Metal Sales (Twin Falls). Get an itemized quote with the full trim package.
- **Wet-set brackets:** usually ship with the laminated column order.
- **14′ × 12′ overhead door:** a local overhead-door dealer, with an installed quote and low-headroom track.
- **Insulated chimney:** a fireplace specialty shop (Idaho Hearth).
- **Ready-mix concrete:** the closest plant to Prairie (CalPortland Boise, Idaho Concrete in Mountain Home).
- **Snow guards:** direct from the manufacturer (SnowTrapper, S-5!).
- **Crushed gravel:** a local pit, much cheaper than bagged.

Quote-required items: do these first

1. Trusses (engineered and stamped)
2. The laminated column package and brackets
3. The steel roof, wall, and trim package
4. The 14′ × 12′ overhead door (installed)
5. Ready-mix concrete (per-yard price and short-load fees)

These five items determine about half of the project cost. Get them quoted before you swing a hammer.

Cost-cut levers (if the budget is tight)

- Site-build the posts instead of buying them laminated ($1,200 saved)
- Skip the interior finish for now ($1,800 saved)
- Skip the insulation and install it when needed ($3,000 saved)
- Skip the mini-split and run the wood stove only ($2,500 saved)
- A 3070 walk door without windows, or lower grade ($300 saved)
- Fewer windows: the 7 windows with their flashing and trim are about $1,900 to $3,500 together
- A standard (non-insulated) wind-rated overhead door ($800 saved, but you will regret it in winter)
- A single-color steel package ($300 saved)
- PT plywood instead of 5/4 PT decking ($200 saved)
- Defer the landing deck to a later phase ($1,800 saved)

Do NOT cut these

- **Engineered trusses with a PE stamp.** 120 psf snow is non-negotiable.
- **Wet-set brackets.** Direct-burial posts rot in 15–25 years.
- **Rainscreen furring.** The whole reason for OSB behind the steel is wasted without it.
- **Tyvek, not Typar or a cheap generic.** Commercial-grade WRB lasts 50 years.
- **Ice and water shield on the full lower slope.** Failure here puts water in the loft.
- **The 2×10 PT splash board.** Do not substitute a 2×8 or skip it.
- **4″ rigid foam under the slab** (or at least 2″ XPS). Without it the slab will sweat.
- **A wind-rated overhead door.** Very few off-the-shelf doors meet 115 mph.

Contingency and Total Project Budget

Line	Low	Midpoint	High
Materials (summed from the line items above)	$79,344	$101,076	$122,808
+ Sales tax (Idaho 6%)	$4,761	$6,065	$7,368
+ Permits, plan review, engineering County permit, the Idaho engineer's stamp on the plans and the trusses, and the electrical permit.	$1,500	$2,500	$4,000
+ Pier holes, compaction testing, and any extra dirt work The pad itself is already built (imported fill), so this is mostly drilling the pier holes and testing the fill.	$1,500	$3,000	$5,000
+ Utility hookup (power feeder) The specs call for no plumbing, so this is the feeder trench and the service connection; the cable and conduit are in phase 18. Wildly site-dependent; the feeder source is not settled.	$1,500	$4,000	$10,000
+ Equipment rental (lift, telehandler, concrete tools) About $1,500–4,000 for a scissor lift and a man-lift on truss-set day.	$1,500	$3,000	$4,500
+ Contingency (15% of everything above)	$13,516	$17,946	$23,051
OWNER-BUILD ALL-IN BUDGET RANGE	**$103,620**	**$137,587**	**$176,728**

Reality check: plan on the **Midpoint** column (about $137,600) as your real budget, and on the **High** column (about $176,700) as the number that makes you stop and reassess. Material costs are volatile: lumber alone can swing 30% in one quarter. Locking in the steel and truss prices early protects about half of the spend.

Length and Cost

Each foot of length

- **Each foot adds about $940 of materials** at the midpoint, about $1,100 all in. The trusses, the steel, the concrete, the foam, the wrap, and the ice-and-water shield grow with the length; the doors, the stair, the deck, and the hoist do not, so a longer building costs less per square foot. Materials come to about $66 per sq ft of floor.
- **The estimate takes no credit for less waste.** The length is a whole number of 8 ft bays, so the posts, the 16 ft girts, the 4×8 sheets, and the 36 in steel panels land on a clean module and need little trimming. Every quantity here scales with the length, so the real cost is probably a few hundred dollars lower. Ask the steel and truss suppliers to quote the package at the length you build.

Mountain Prairie Acres · Workshop pole barn materials and cost · updated October 23, 2026 · prices are 2026 ballpark figures and will drift; get quotes before ordering.